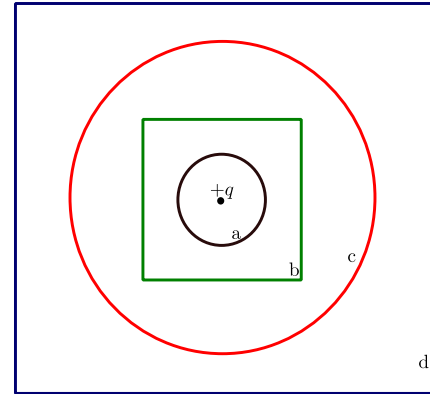


Part-1, Conceptual Questions (Total 20 points)

Show your work!

- 1) Figure shows in cross section, two Gaussian spheres and two Gaussian cubes that are centered on a positively charged particle.

(a) (4 points) Rank the net flux through the four Gaussian surfaces, greatest first.



(b) (4 points) Rank the magnitudes of the electric fields on the surfaces, greatest first, and indicate whether the magnitudes are uniform or variable on each surface.

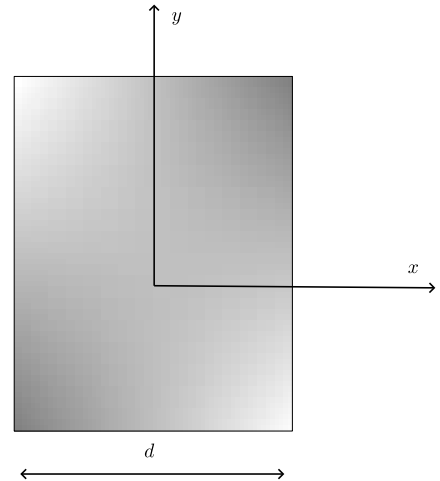
- 2) (3 points) The problems solvable by Gauss' law fall into three categories, according to their symmetry: spherical, cylindrical, and planar. Give some examples for each category.
- 3) (3 points) When an electric current is flowing through a wire connected between a source and sink of electric charge - such as terminals of the battery - there is an electric field inside the wire, even though wire is a conductor. Why does our conclusion about zero electric field inside a conductor not apply to this case?
- 4) (3 points) If a high voltage power cable falls on top of your automobile, you will probably be safest if you remain inside the automobile. Why?
- 5) (3 points) Explain what conservative force means and give an example.

Part-2, Problems (Total 100 points)

Problem 1 (25 points)

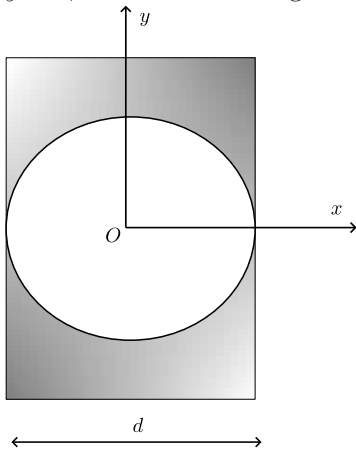
A flat slab of nonconducting material has a volume charge density ρ . The slab has thickness d which is small compared to the dimensions of the slab. Express your answers in terms of some or all of the given quantities and appropriate constants.

(i) (5 points) Find the electric field (both magnitude and direction) inside the slab.



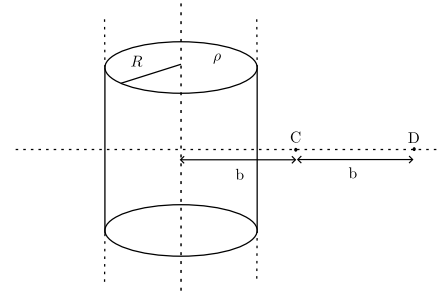
(ii) (5 points) Find the electric field (both magnitude and direction) outside the slab.

(iii) (15 points) Assume that a spherical cavity of radius $d/2$ is cut out and left empty such that its origin is at $x = 0$ and $y = 0$, as shown in the figure. Find the electric field (both magnitude and direction) at any point on the x axis.



Problem 2 (25 points)

An infinitely long nonconducting cylinder of radius R and uniform positive volume charge density ρ , and it is in a space with zero gravity. The cylinder's axis lies in the xy plane, as shown in the figure. The points C and D are at radial distances b and $2b$, respectively, from the axis of the cylinder. Express your answers in terms of some or all of the given quantities and related constants as needed.



(i) (5 points) Using Gauss' law, find the magnitude of the electric field $E(r)$ outside the cylinder at a radial distance $r > R$.

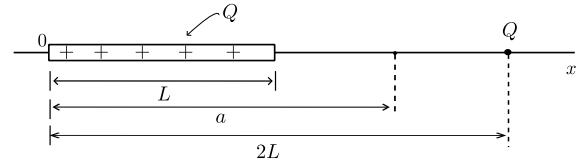
(ii) (5 points) Using Gauss' law, find the electric field $E(r)$ inside the cylinder $r < R$. Indicate the direction of the electric field.

(iii) (10 points) Find the electric potential difference between $V_C - V_D$ between points C and D .

(iv) (5 points) Find the work W_{app} that you must do to carry an electron from point C to point D without changing its kinetic energy.

Problem 3 (25 points)

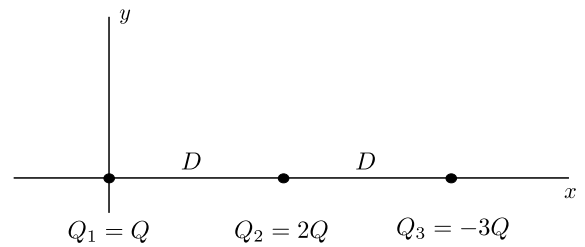
A thin, nonconducting rod of length L has a positive charge Q uniformly distributed along its length. It lies along the x axis between points $x = 0$ and $x = L$. A point-like object with identical positive charge Q lies on the x axis at the points $x = 2L$. Express your answers in terms of $k = (1/4\pi\epsilon_0)$, Q and L as needed. (Hint: integral gets simpler if you change the integral variable from x to $u = a - x$.)



- (i) (15 points) Let $x = a$ be a point on the x axis between the two objects where the electric field $\vec{E}$ is zero. Find a .
- (ii) (10 points) Relative to $V = 0$ at $x = \infty$, determine the electric potential V at the point $x = a$.

Problem 4 (25 points)

Three point charges Q_1 , Q_2 and Q_3 are placed on the x axis, as shown in the figure. The central charge Q_2 is a distance D from each of the others. Take $Q_1 = Q$, $Q_2 = 2Q$, and $Q_3 = -3Q$. Express your answer in terms of π , ϵ_0 , Q , D as needed.



(i) (12 points) Find the net force $\vec{F}$ on the point charge Q_2 . Express your answers in terms of unit vectors.

(ii) (13 points) Find the electrostatic potential energy U of the system of three point charges.